Candidate Problem Set — Steady Currents, Static Magnetic Fields, Inductance

Nine problems, three per section, mid-level with spread

This short problem set collects nine candidate problems organized into three sections, with one problem from each of three sources: the original five (P.4-10, P.4-11, P.5-3, P.5-13, P.5-16), the course homework PDFs, and Nayfeh & Brussel's *Electricity and Magnetism*. Within each section the difficulty scales from ⋆ ⋆⋆ (easier mid) through ⋆⋆⋆ (harder mid).

Steady Currents & Resistance

1.1 — Curved-sides resistance of a quarter washer

Source: Cheng P.4-11 (from original-5 pool) *Difficulty: ★★△ (easier mid)*

Refer to the flat conducting quarter-circular washer of Example 4-4: inner radius a, outer radius b, axial thickness h, uniform conductivity σ, angular extent $\pi/2$. Find the resistance between the two *curved* sides (the inner cylindrical face at $r = a$ and the outer cylindrical face at $r = b$).

Hint: the current flows radially, so the area available for conduction varies with r.

1.2 — Parallel-plate capacitor with two lossy dielectrics

Source: HW4, problem P.4-8 *Difficulty: ★★△ (mid)*

A d-c voltage V_0 is applied across a parallel-plate capacitor of area S. The space between the conducting plates is filled with two different lossy dielectrics of thicknesses d_1 and d_2, permittivities ϵ_1 and ϵ_2, and conductivities σ_1 and σ_2, stacked in series (layer 1 adjacent to terminal a, layer 2 adjacent to terminal b). Determine:

(a) the current density $\mathbf{J}$ between the plates (steady state),

(b) the electric field intensities $\mathbf{E}_1$ and $\mathbf{E}_2$ in the two dielectrics, and

(c) the equivalent R–C circuit seen between terminals a and b.

Hint: in steady state, $\nabla \cdot \mathbf{J} = 0$ forces $\mathbf{J}$ to be the same in both layers — this determines how the voltage divides.

1.3 — Cube with position-dependent conductivity

Source: Nayfeh 7.7 (book) *Difficulty: ★★★ (harder mid)*

A block of conducting material is in the shape of a cube of side a. Its conductivity is not uniform: at any point in the block,

$$\sigma(x) \;=\; \alpha\,(a + x),$$

where α is a constant. Assume current flows only along the x-axis from face S (at $x = 0$) to the opposite face S' (at $x = a$).

(a) Taking the electrostatic potential Φ to depend only on x, use $\boldsymbol{\nabla}\cdot\mathbf{J}=0$ with $\mathbf{J}=\sigma\mathbf{E}$ to show that

$$\frac{\partial}{\partial x}\left[(a+x)\,\frac{\partial\Phi}{\partial x}\right]=0.$$

(b) Solve this equation and show that the potential difference between S and S' has the form $\Delta\Phi = A\ln 2$, where A is an integration constant.

(c) Compute the total current flowing through the cube, and show that the resistance between S and S' is

$$R \;=\; \frac{\ln 2}{\alpha a^2}.$$

Static Magnetic Fields

2.1 — B of a finite straight wire

Source: Cheng P.5-3 (from original-5 pool) *Difficulty: ★★△ (easier mid)*

A direct current I flows in a straight filamentary conductor from point P_1 to point P_2 (as in Fig. 5-24).

(a) Prove that $\mathbf{B}$ at an observation point P, specified by the perpendicular distance r from the wire and by the two angles α_1 and α_2, is

$$\mathbf{B}_P \;=\; \hat{\mathbf{a}}_\phi\,\frac{\mu I}{4\pi r}\,(\sin\alpha_2-\sin\alpha_1).$$

(b) Verify that this reduces to the familiar result $\mathbf{B}_P=\hat{\mathbf{a}}_\phi\,\mu I/(2\pi r)$ for an infinitely long wire.

2.2 — Coaxial cable: B in every region

Source: HW5, problem P.5-6 *Difficulty: ★★△ (mid)*

A current I flows in the inner conductor of an infinitely long coaxial line and returns via the outer conductor. The inner conductor has radius a; the outer tubular conductor has inner and outer radii b and c, respectively (so $a<b<c$). Assume both conductors are solid cylinders with uniformly distributed currents. Find the magnetic flux density $\mathbf{B}$ in *all four* regions

$$0<r<a,\quad a<r<b,\quad b<r<c,\quad r>c,$$

and sketch $|\mathbf{B}|$ versus r.

Hint: Ampère's law with cylindrical symmetry; in region $b<r<c$ only a fraction *of the return current has been enclosed.*

2.3 — Hairpin wire bent around a cylinder

Source: Nayfeh 8.14 (book) *Difficulty: ★★★ (harder mid)*

A very long straight wire is bent into a hairpin: the wire comes in along a straight path, wraps 180° around the outside of a wooden cylinder of radius b, and exits along a parallel straight path

on the opposite side of the cylinder (so the wire's two straight halves are anti-parallel, separated by $2b$, and joined by a semicircular arc of radius b). A current I flows around this whole path.

Find $\mathbf{B}$ at the *center* of the cylinder (in the plane of the wire).

Hint: superpose three contributions — two semi-infinite straight segments plus one 180° *circular arc. Use the finite-wire formula from problem 2.1.*

Inductance

3.1 — Mutual inductance: long wire + equilateral triangle

Source: Cheng P.5-16 (from original-5 pool) *Difficulty: ★★△ (easier mid)*

Determine the mutual inductance between a very long straight wire and a coplanar conducting equilateral triangular loop of side b. The vertex of the triangle nearest the wire is at perpendicular distance d, with the opposite side of length b parallel to the wire (see Fig. 5-27).

Hint: integrate $\mathbf{B}\cdot d\mathbf{A}$ *using vertical strips whose height varies linearly with the horizontal distance from the near vertex.*

3.2 — Self-inductance of a toroidal coil (circular cross section)

Source: HW5, problem P.5-15 *Difficulty: ★★△ (mid)*

Determine the self-inductance of a toroidal coil of N turns of wire wound on an air frame with mean radius r_o and a *circular* cross section of radius b. Obtain an approximate closed-form expression for L in the limit $b \ll r_o$.

Hint: for $b \ll r_o$, $\mathbf{B}$ *is nearly uniform across the cross section (equal to its value at the mean radius), so the flux integral simplifies dramatically.*

3.3 — Small coil inside a solenoid: induced emf and current

Source: Nayfeh 11.2 (book) *Difficulty: ★★★ (harder mid)*

A long solenoid has $n = 100$ turns/cm and a diameter of 3.0 cm. A thin coil of $N_2 = 20$ turns and diameter 2.0 cm is placed *inside* the solenoid with its axis parallel to the solenoid's axis. The solenoid carries a time-varying current

$$I(t) \;=\; 3t + 2t^2 \quad \text{(A, with } t \text{ in s)}.$$

(a) Determine the induced emf $\varepsilon(t)$ in the small coil.

(b) If the small coil has resistance $R = 0.15\ \Omega$, determine the instantaneous current in it at $t = 2$ s. (Neglect the flux produced by the induced current itself.)

Hint: the solenoid's $\mathbf{B}$ *is uniform inside and threads every turn of the small coil, so the flux linkage is* $\lambda = N_2\, B_{sol}\, A_{coil}$. *Faraday's law then gives* $\varepsilon = -d\lambda/dt$.

How to use this sheet. Each section is self-contained and can be attempted independently. Within a section, problem 1 fixes basic technique, problem 2 applies it to a standard device/geometry, and problem 3 pushes a little further — either by breaking symmetry (Section 1), stacking multiple

sources (Section 2), or adding a time dependence (Section 3). If you pick one from each row, you get a balanced *three-problem* assignment covering all three topics at mixed difficulty; if you pick a whole column (same difficulty across all three sections), you get a rung of increasing sophistication.